MINING DEALER BEST PRACTICE

Backlog Runtime Optimization for Mining Equipments

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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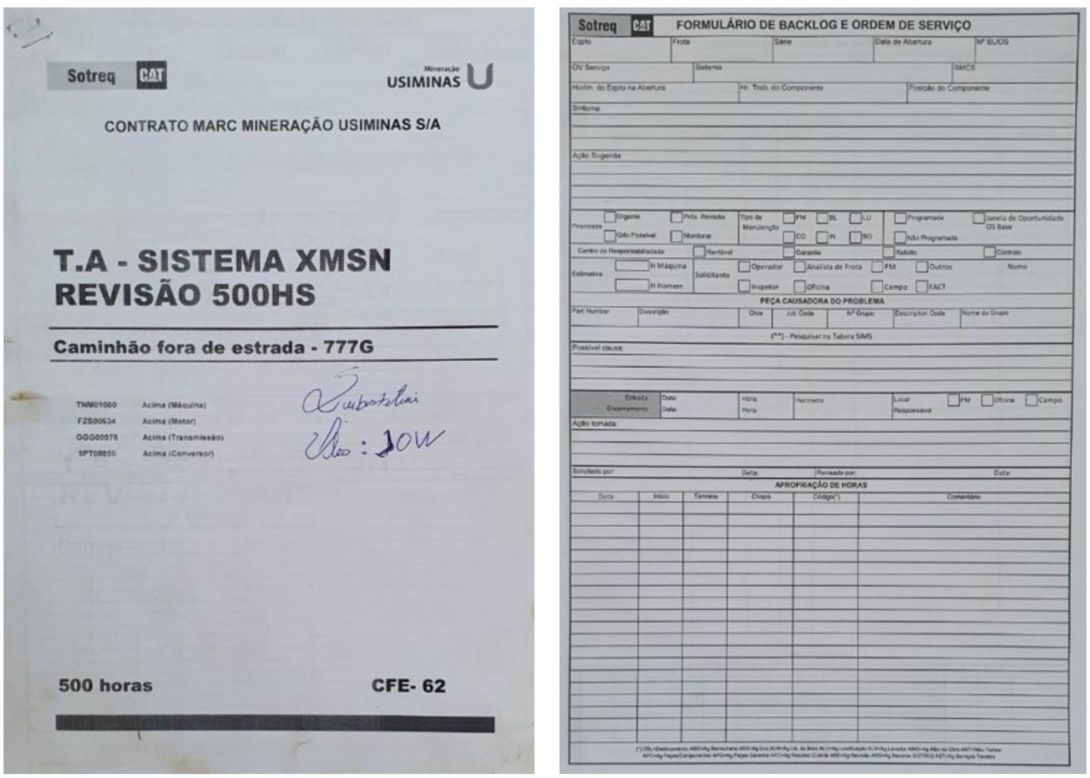
1.0 Introduction

Facing mining sector current scenario, customers have been seeking partnerships with dealers, searching for best practices which allow them to reduce equipment's operating cost, making productivity more efficient, and thus allowing their process to be more competitive when compared to competitors.

During maintenance activities, it has been observed that many technicians stop their activities and take a considerable time to search for literature on procedures, print and interpret, since most of the contract technician come from Sotreq's mechanical school and still do not have the ability to find literature quickly, the time to execute preventive measures is significant.

When performing preventive maintenance services, such as the 500-hour interval for instance, technicians receive a maintenance plan form, called Backlog and Service Order Form, in which the tasks are qualified as pending execution, as shown in the following Picture 01.



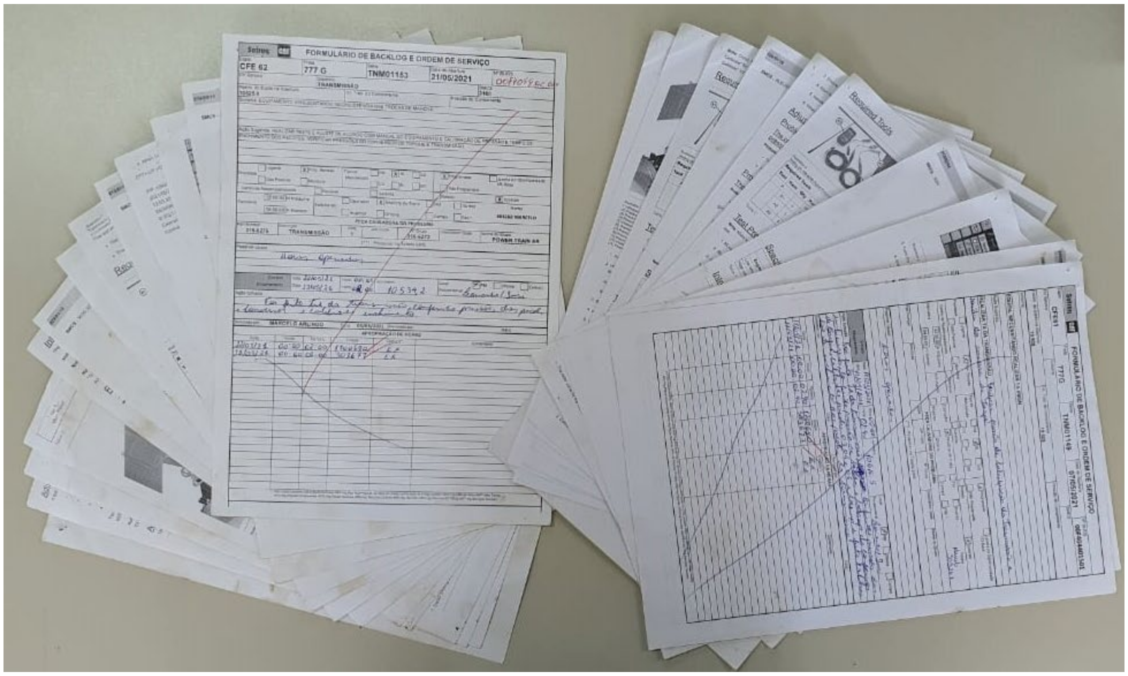
Sotreq CAT
Mineração USIMINAS
CONTRATO MARC MINERAÇÃO USIMINAS S/A
T.A - SISTEMA XMSN
REVISÃO 500HS
Caminhão fora de estrada - 777G
TNM01000 Acima (Máquina)
FZS00634 Acima (Motor)
GGG00978 Acima (Transmissão)
SPT08850 Acima (Conversor)
500 horas CFE- 62

Sotreq CAT FORMULÁRIO DE BACKLOG E ORDEM DE SERVIÇO
Eqpto | Frota | Série | Data de Abertura | Nº BL/OS
OV Serviço | Sistema | SMCS
Horím. do Eqpto na Abertura | Hr. Trab. do Componente | Posição do Componente
Sintoma:
Ação Sugerida
Prioridade: Urgente, Próx. Revisão, Qdo Possível, Monitorar | Tipo de Manutenção: PM, BL, LU, CO, IN, BO | Programada, Não Programada | Janela de Oportunidade OS Base
Centro de Responsabilidade: Rentável, Garantia, Refeito, Contrato
Estimativa: H Máquina, H Homem | Solicitante: Operador, Analista de Frota, PM, Outros, Inspetor, Oficina, Campo, FACT | Nome
PEÇA CAUSADORA DO PROBLEMA
Part Number | Descrição | Qtde | Job Code | Nº Grupo | Description Code | Nome do Grupo
(**) - Pesquisar na Tabela SIMS
Possível causa:
Entrada / Encerramento: Data | Hora | Horímetro | Local / Responsável | PM | Oficina | Campo
Ação tomada:
Solicitado por | Data | Revisado por | Data
APROPRIAÇÃO DE HORAS
Data | Início | Término | Chapa | Código(*) | Comentário

Picture 01 – Example of a Backlog and Service Order Form.

Aiming to accomplish those tasks in accordance with Caterpillar quality standards, technicians must be oriented by literature guidance's. Thus, they took some time searching and printing them and once the service is completed, those are discarded.

To reduce the field technician's search time, service technician becomes responsible for searching, printing, and delivering the procedures to the field technicians as well as the backlog forms to be carried out according to Picture 02:



Picture 02 – 500-hour preventive maintenance with a transmission test and adjustment.
Approximately 15 printed sheets.

However, it was noticed the process could still be improved, since it was resulting in many prints and several paper sheets per activity, prejudicing the environment.

Observing the necessity to optimize backlogs runtime, this project was conceived and executed.

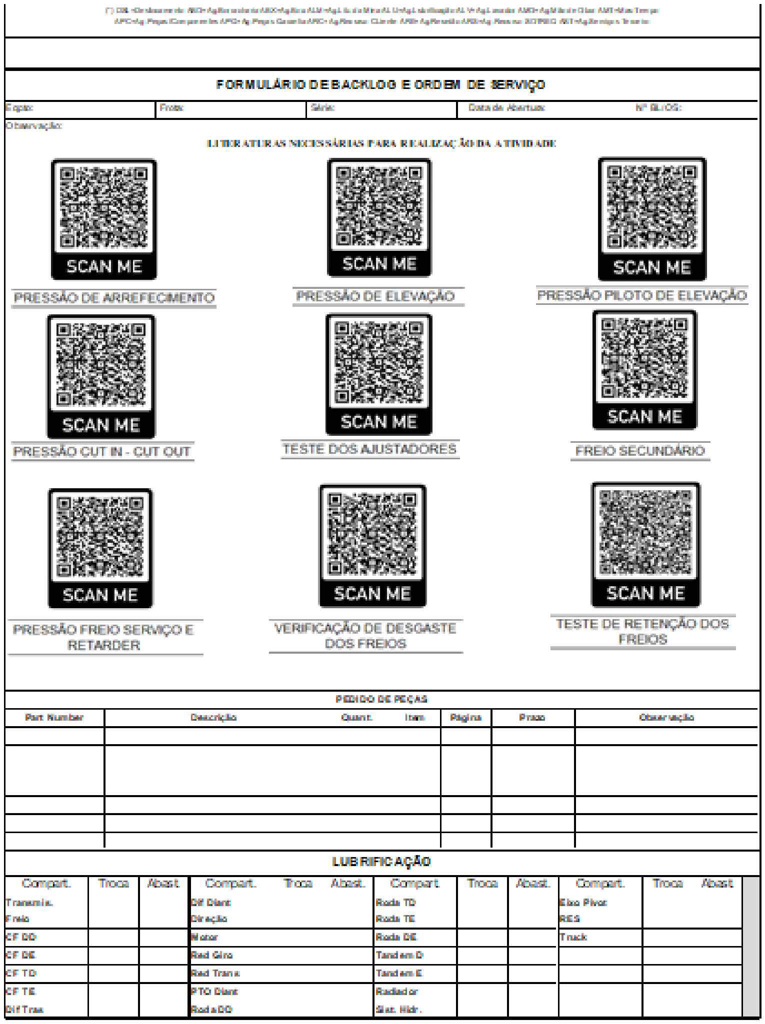
2.0 Best Practice Description

All procedures printed before, are generated via QR code. Only these codes are printed and added to the backlog sheet. In this way, technicians, using the corporate cell phones, scan the code and the procedures are visualized on its screen, encouraging sustainability, and saving time to more critical activities.

Two studies were carried out to verify the execution time of a 500-hour preventive service:

1. The first study measured the time spent by the field team to gather the necessary information, print the procedures, and perform the service as it was originally done. Considering two technicians running, on average, it used to take about 16-hour-service plus 2 hours for the technician finding the necessary procedures in the system, read and interpret. In addition, 5 to 15 sheets were printed.

2. In the second, the service technicians started to check in advance the necessary literature to perform the task and create a QR code that directs to the SIS2 link. These codes are printed on the back page of the backlog and sent to the area for technicians to run as Picture 03 illustrates.



FORMULÁRIO DE BACKLOG E ORDEM DE SERVIÇO

Eqpto:	Frota:	Série:	Data de Abertura:	Nº BLOGS:

Observação:

LITERATURAS NECESSÁRIAS PARA REALIZAÇÃO DA ATIVIDADE

PEDIDO DE PEÇAS

Part Number	Descrição	Quant.	Item	Página	Prazo	Observação

LUBRIFICAÇÃO

Compart.	Troca	Abast.	Compart.	Troca	Abast.	Compart.	Troca	Abast.	Compart.	Troca	Abast.
Transmis.			Dif Diant			Roda TD			Eixo Pivot		
Freio			Direção			Roda TE			RES		
CF DD			Motor			Roda DE			Truck		
CF DE			Red Giro			Tandem D					
CF TD			Red Trans			Tandem E					
CF TE			PTO Diant			Radiador					
Dif Tras			Roda DD			Sist. Hidr.					

Picture 03 – Example of an optimized backlog sheet containing QR codes.

After this implementation, the time for a 500-hour preventive service decreased around two hours. In addition, the printed backlog reduced to two pages.
In 2020, for instance, 135 revisions were carried out in the operational fleet at Usiminas Mineração, containing five D8T track-type tractors. Considering each review approximately 15 sheets are printed, 2025 sheets were saved by year.

After the project implementation, 270 hours and 270 sheets are saved yearly on preventive service currently.

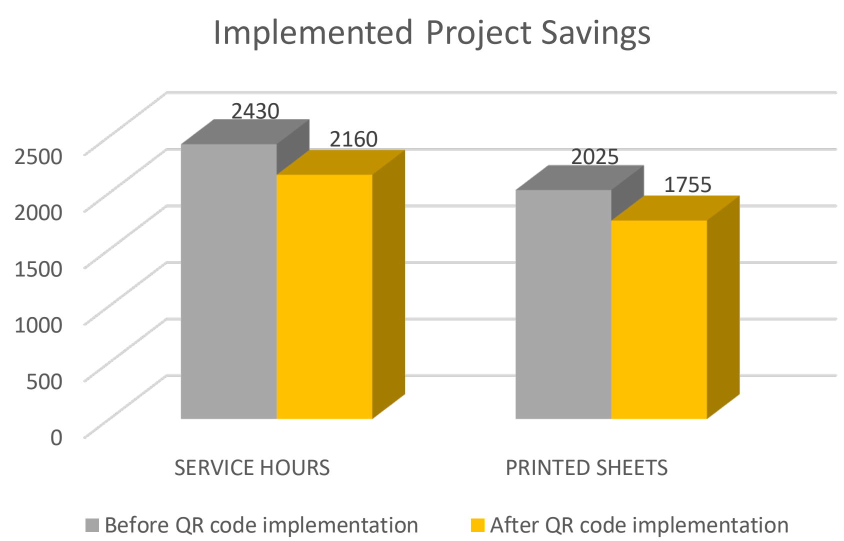
3.0 Implementation Steps

1. Measure execution time on a routine service maintenance,
2. Define time interval for each activity related to the preventive service, such as literature search and print,
3. Anticipate and catalog all literature related to backlog activities before field technicians,

4. Generate QR code via online program containing all necessary literature to perform backlog activity,
5. Print a single sheet containing backlog tasks and in the back of that sheet print generated QR codes,
6. Share that sheet to field technicians, who should use their corporate cellphone to read the code and perform the activities correctly.

4.0 Benefits

- Environmentally friendly process
- Reduction from 2430 to 2160 hours in 500-hour preventive maintenance, which stands for 11%. Also decrease of 13% print material according to Graphic 01 considering a year period:



Graphic 01 – Savings related to QR code implementation.

*For this test 500-hour preventive maintenance was elected, however it can expand to different services.

5.0 Resources Required

- Printer
- QR code generation program.

6.0 Supporting Attachments / References

See https://br.qr-code-generator.com for QR code generation.

7.0 Related Best Practices

N.A

8.0 Acknowledgements

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